

TASK 9: Backing up and recovery in databases CO4, K3

Perform following backup and recovery scenarios.

- a) Recovering a NOARCHIVELOG Database with Incremental Backups
- b) Restoring the Server Parameter File
- c) Performing Recovery with a Backup Control File

Scenario 1: Recovering a NOARCHIVELOG Database with Incremental Backups

-- Step 1: Backup Database

```
BACKUP DATABASE [database_name] TO DISK = 'backup_file.bak' WITH  
NOFORMAT, NOINIT, NAME = 'Full Database Backup', SKIP, REWIND,  
NOUNLOAD, STATS = 10
```

-- Step 2: Create Incremental Backup

```
BACKUP DATABASE [database_name] TO DISK = 'incremental_backup.bak' WITH  
DIFFERENTIAL, NOFORMAT, NOINIT, NAME = 'Incremental Database Backup',  
SKIP, REWIND, NOUNLOAD, STATS = 10
```

-- Step 3: Simulate Data Loss

-- Intentionally delete or modify data.

-- Step 4: Restore Database

```
RESTORE DATABASE [database_name] FROM DISK = 'backup_file.bak' WITH  
REPLACE
```

-- Step 5: Apply Incremental Backup

```
RESTORE DATABASE [database_name] FROM DISK = 'incremental_backup.bak'  
WITH REPLACE
```

-- Step 6: Recover Database

```
RECOVER DATABASE [database_name]
```

```
-- Step 7: Open Database
```

```
ALTER DATABASE [database_name] SET ONLINE
```

Scenario 2: Restoring the Server Parameter File (SPFILE)

```
-- Step 1: Backup SPFILE
```

```
BACKUP SERVER PARAMETER FILE TO FILE = 'spfile.bak';
```

```
-- Step 2: Simulate SPFILE Loss
```

```
-- Delete or modify SPFILE.
```

```
-- Step 3: Restore SPFILE
```

```
STARTUP MOUNT
```

```
RESTORE SERVER PARAMETER FILE FROM FILE = 'spfile.bak';
```

```
SHUTDOWN
```

```
STARTUP
```

Scenario 3: Performing Recovery with a Backup Control File

```
-- Step 1: Backup Control File
```

```
BACKUP CONTROLFILE TO FILE = 'controlfile.bak';
```

```
-- Step 2: Simulate Control File Loss
```

```
-- Delete or modify control file.
```

```
-- Step 3: Restore Control File
```

```
STARTUP MOUNT
```

```
RESTORE CONTROLFILE FROM FILE = 'controlfile.bak';
```

ALTER CONTROLFILE REUSE;

-- Step 4: Recover Database

RECOVER DATABASE USING BACKUP CONTROLFILE;

-- Step 5: Open Database

ALTER DATABASE OPEN RESETLOGS;

SQL Server Commands:

- BACKUP DATABASE
- RESTORE DATABASE
- RECOVER DATABASE
- ALTER DATABASE
- BACKUP SERVER PARAMETER FILE
- RESTORE SERVER PARAMETER FILE
- BACKUP CONTROLFILE
- RESTORE CONTROLFILE

Result: